10.1 Lines and Segments That Intersect Circles

Name _____ Period ____ Date _____

Geometry 10.1 · Student sheet

I can…

- name the special segments and lines of a circle.
- use the fact that a tangent is perpendicular to the radius at the point of tangency.
- use the fact that two tangent segments from the same point are congruent.

Vocabulary · fill in the missing word

Chord — a segment whose _____ are on a circle

Secant — a line that _____ a circle in two points

Tangent — a line in the plane of a circle that _____ the circle in exactly one point

Point of tangency — the point where a tangent line _____ the circle

Part 1 · Discovery: lines that touch a circle

Draw a circle and one radius. Through the radius's endpoint on the circle, draw the line that touches the circle **only once**. From a point outside the circle, draw the two lines that just touch the circle.

What I measured	Measurement
Angle between the radius and the touching line	
First segment from the outside point	
Second segment from the outside point	

What do you notice?

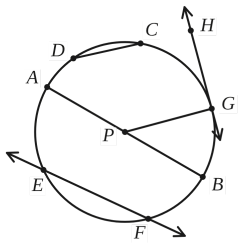
Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

In circle P, name each of the following.

a. a chord b. a diameter c. a radius

d. a secant e. a tangent f. a point of tangency

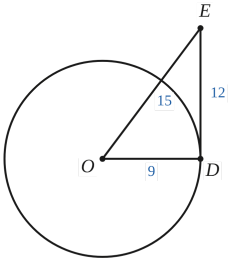


Tangent Line to Circle Theorem: a line is tangent to a circle if and only if it is _____ to a radius at its endpoint on the circle.

Testing for a tangent: use the Converse of the _____ Theorem. If $r^2 + (\text{segment})^2 = d^2$, the line is tangent.

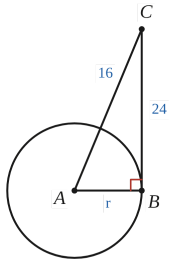
Example 2

Is $\overline{ED}$ tangent to circle O? Explain.



Example 3

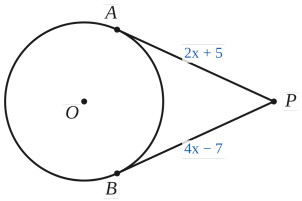
$\overline{BC}$ is tangent to circle A at B. Find the radius r.



External Tangent Congruence Theorem: tangent segments from a common external point are _____.

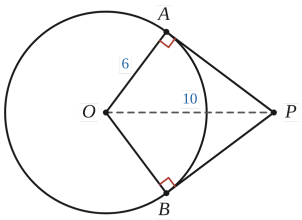
Example 4

$\overline{PA}$ and $\overline{PB}$ are tangent to circle O. Find the value of x. Then find PA.



Example 5

$\overline{PA}$ and $\overline{PB}$ are tangent to circle O. The radius is 6 and OP = 10. Find PA, and the perimeter of quadrilateral OAPB.



Part 3 · Partner practice

Partner 1

A circle has radius 7. A segment from a point on the circle is 10 long, and its far end is 12 from the center. Is the segment tangent to the circle?

Partner 2

Two tangent segments from the same point have lengths 3x + 2 and 5x − 8. Find x and the length of each segment.